Biochemistry Cumulative Exam
January 23, 2015

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid aspartate, but no carbohydrate, or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (40) Outline a metabolic pathway for the mouse to synthesize glyceraldehyde-3-phosphate on this diet using aspartate as the sole source of carbon, nitrogen and energy. How many aspartate molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of glyceraldehyde-3-phosphate? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, and CO_2 generated.

2. (40) Outline a metabolic pathway for the mouse to synthesize succinate on this diet using aspartate as the sole source of carbon, nitrogen and energy. How many aspartate molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of succinate? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, and CO_2 generated.

3. (20) If aspartate is labeled with ^{14}C at carbon 2, which carbon(s) of glyceraldehyde-3-phosphate and succinate will first become labeled?

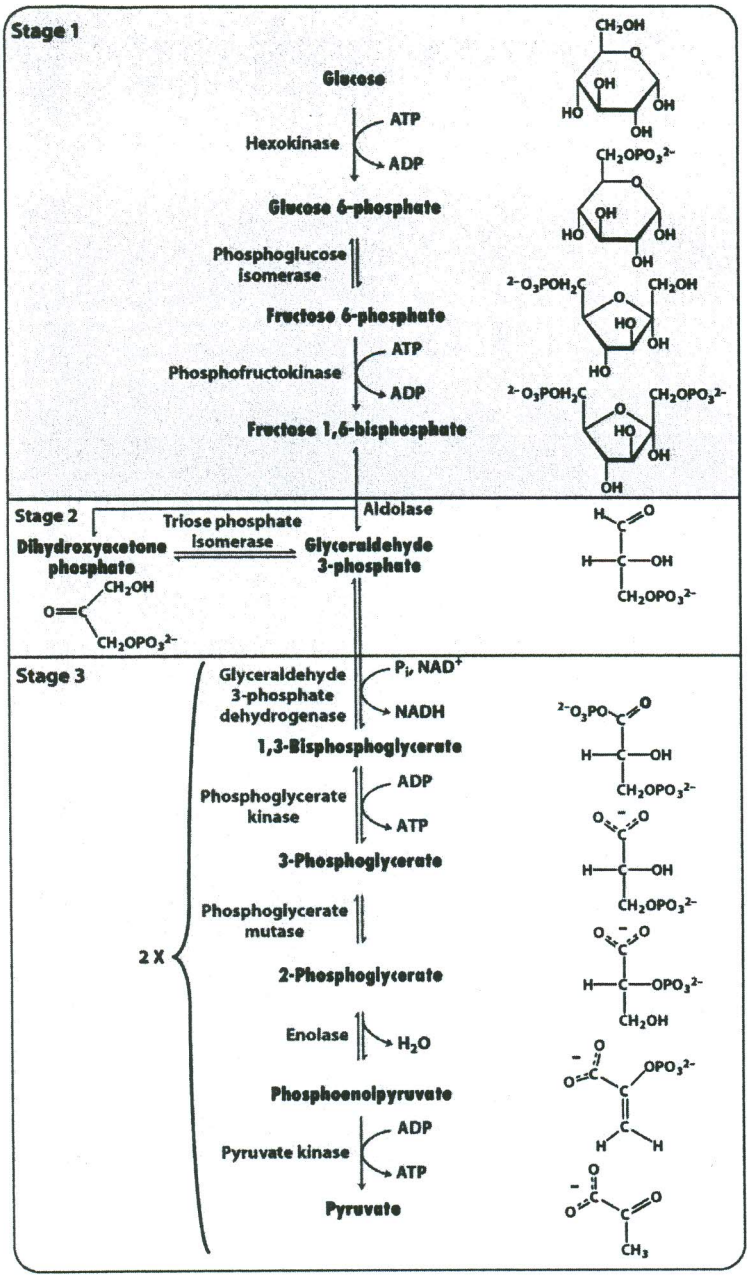

Figure 16-2
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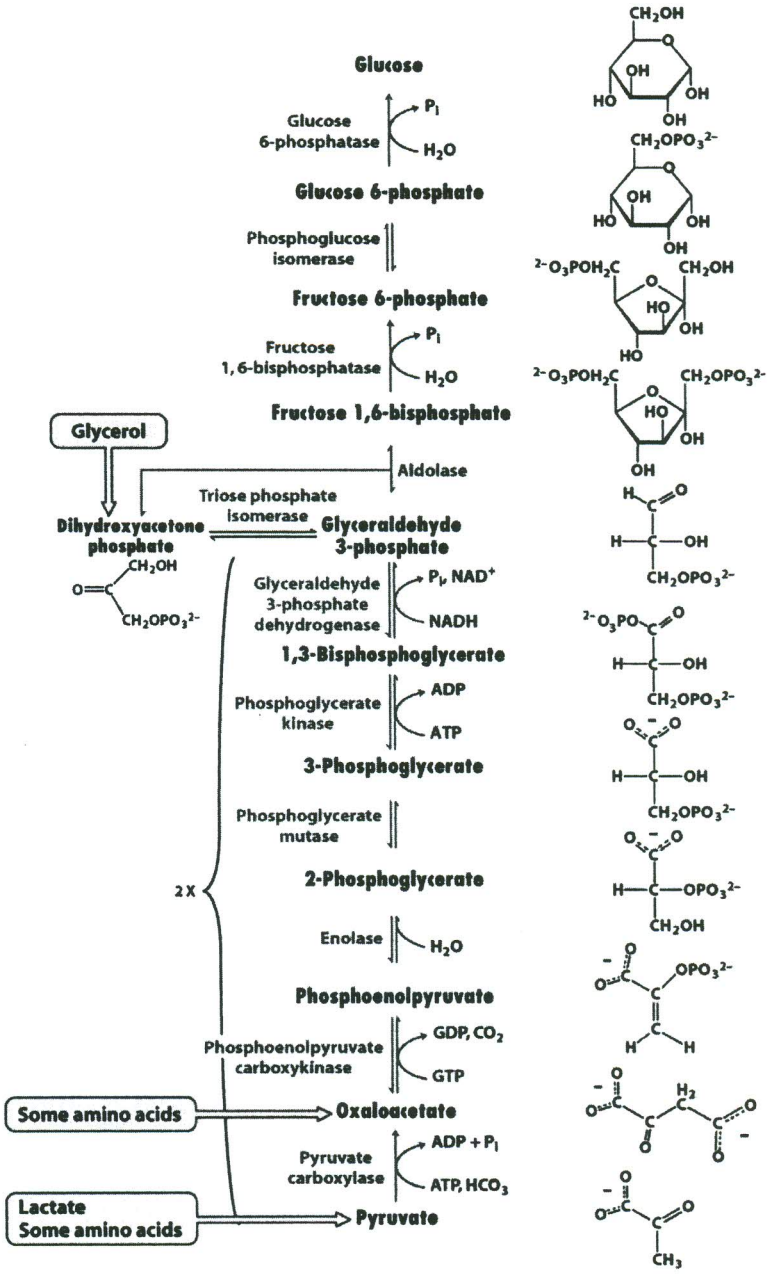

Figure 16-22
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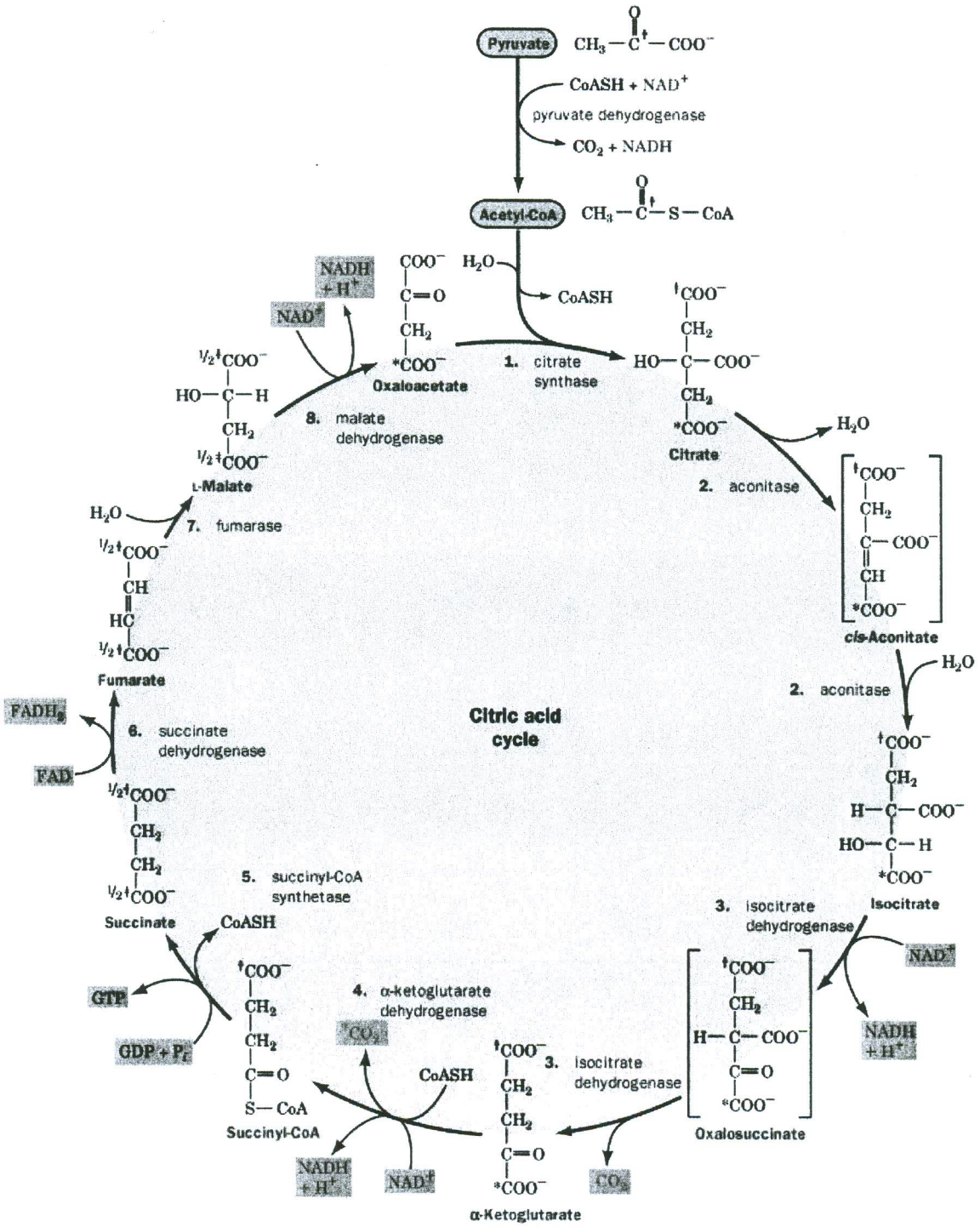
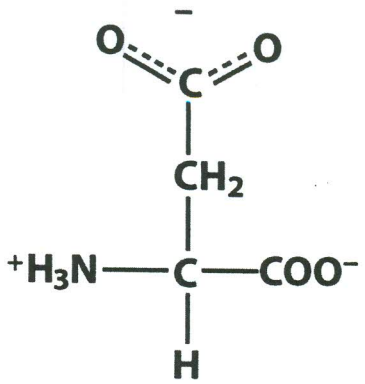
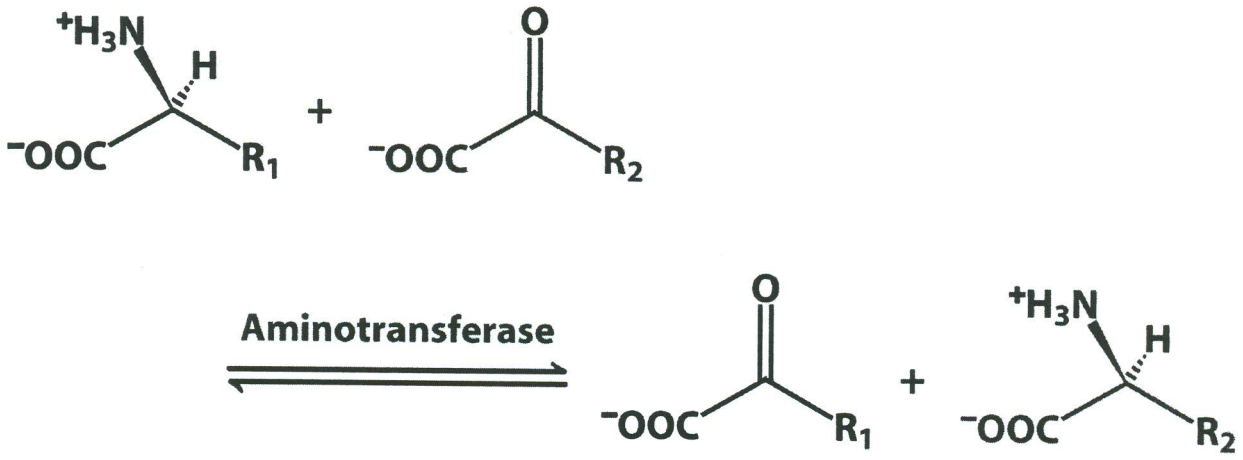

Pyruvate
CoASH + NAD+
pyruvate dehydrogenase
CO2 + NADH
Acetyl-CoA
CoASH
H2O
1. citrate synthase
Oxaloacetate
Citrate
2. aconitase
cis-Aconitate
H2O
2. aconitase
Isocitrate
3. isocitrate dehydrogenase
NAD+
NADH + H+
Oxalosuccinate
3. isocitrate dehydrogenase
CO2
α-Ketoglutarate
4. α-ketoglutarate dehydrogenase
CO2
CoASH
NADH + H+
NAD+
Succinyl-CoA
5. succinyl-CoA synthetase
CoASH
GTP
GDP + Pi
Succinate
6. succinate dehydrogenase
FAD
FADH2
Fumarate
7. fumarase
H2O
L-Malate
8. malate dehydrogenase
NAD+
NADH + H+
Citric acid cycle



Aspartate (Asp, D)



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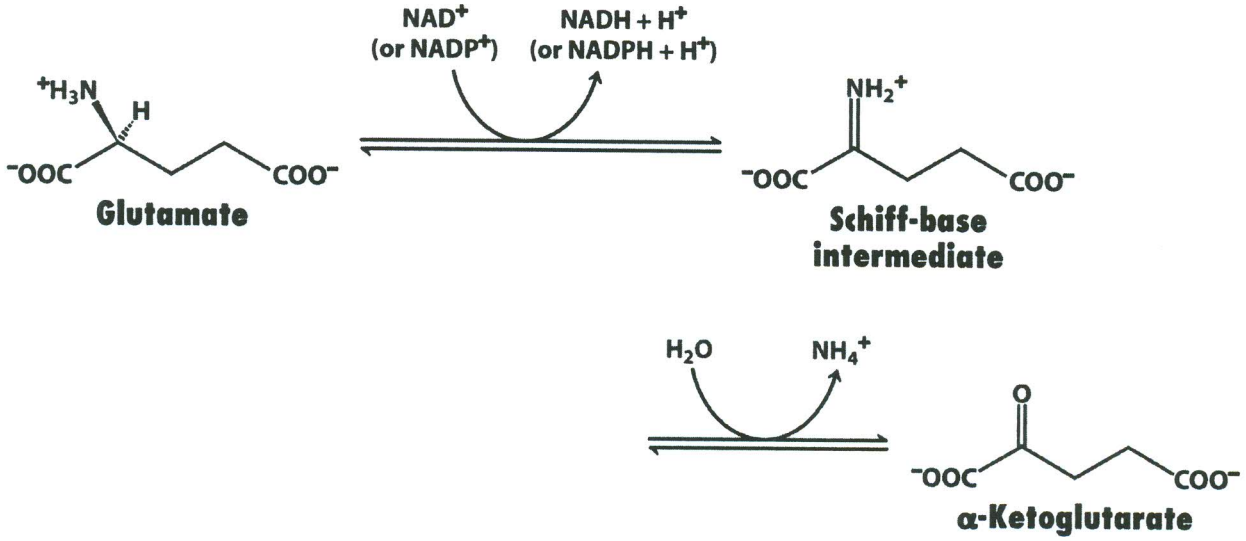

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